

Detailed and Updated Summary:

Robotic Technologies for High-Throughput Plant Phenotyping: Contemporary Reviews and Future Perspectives

NECKER

Based on the original PDF

November 12, 2025

Abstract

Plant phenotyping involves the quantitative and qualitative measurement of morphological, biochemical, physiological, structural, and performance-related traits across different scales of plant organization. It is fundamental for crop improvement programs, especially when facing global challenges such as population growth, climate change, drought, extreme weather events, soil degradation, and water scarcity. These pressures demand a significant increase in the production of food, feed, fiber, fuel, and high-value chemicals. Plant breeders develop high-yielding and stress-tolerant varieties using genomic tools such as DNA sequencing, genotyping, marker-assisted selection, and genomic prediction. However, linking genotypes to complex traits—such as growth, yield, and stress response—requires high-quality phenotypic data collected across thousands of genotypes, multiple traits, and diverse environments.

Contents

1	Introduction and Context	1
2	Robotic Phenotyping Systems	2
2.1	Indoor Systems	2
2.1.1	Non-Contact Systems	2
2.1.2	Contact-Based Systems	2
2.2	Outdoor Systems	2
2.2.1	Non-Contact Unmanned Ground Vehicles (UGVs)	2
2.2.2	Contact-Based UGVs	3
3	Summary of Current Capabilities	3
4	Key Challenges	3
5	Future Advancements and Applications	3
5.1	Sensor-Controller Fusion	3
5.2	Internet of Robotic Things (IoRT)	4
5.3	Renewable Energy	4
5.4	New Measurable Traits	4
5.5	Greenhouse Integration	4
5.6	Swarm Robotics	4
5.7	UAV-UGV Collaboration	4
6	Conclusion	4
7	References	4

1 Introduction and Context

Plant phenotyping involves the quantitative and qualitative measurement of morphological, biochemical, physiological, structural, and performance-related traits across different scales of plant organization. It is fundamental for crop improvement programs, especially when facing global challenges such as population growth, climate change, drought, extreme weather events, soil degradation, and water scarcity. These pressures demand a significant increase in the production of food, feed, fiber, fuel, and high-value chemicals. Plant breeders develop high-yielding and stress-tolerant varieties using genomic tools such as DNA sequencing, genotyping, marker-assisted selection, and genomic prediction. However, linking genotypes to complex traits—such as growth, yield, and stress response—requires high-quality phenotypic data collected across thousands of genotypes, multiple traits, and diverse environments.

Traditional phenotyping methods rely on manual, destructive sampling at specific growth stages. These approaches are labor-intensive, slow, expensive, error-prone, and incapable of tracking individual plants over time. Often, only yield is measured—a trait with low heritability. In contrast, non-destructive, high-throughput phenotyping systems enable repeated and scalable measurements in both controlled and field environments.

Robotic systems represent the next evolution: fully autonomous platforms that integrate sensors, computation, and actuation to perceive plants and their environment, make adaptive decisions, and perform precise measurements or navigation. Unlike conveyor-belt-based systems or simple sensor carriers, true robotic platforms utilize artificial intelligence—such as deep learning, fuzzy logic, and genetic algorithms—to detect crop rows, plant organs, obstacles, and optimal trajectories. This review synthesizes current robotic phenotyping technologies and outlines future directions.

2 Robotic Phenotyping Systems

Robotic platforms are classified by environment: indoor (greenhouses, growth chambers) and outdoor (field conditions).

2.1 Indoor Systems

In controlled environments, plants are either stationary or moved via conveyor belts. Robots interact with them using manipulators.

2.1.1 Non-Contact Systems

Robotic arms equipped with RGB cameras, Time-of-Flight (ToF) sensors, or 3D laser scanners reconstruct plant models to estimate height, volume, and leaf area surface:

- A stationary robotic arm with a ToF camera measures stalk height and leaf length in maize seedlings.
- A 7-degree-of-freedom (DOF) arm with a 3D laser scanner calculates surface area and volume in *Arabidopsis* and barley.
- Occlusion from overlapping leaves limits single-view accuracy. Multi-view imaging helps, but adaptive planning is superior.
- A three-arm system with depth cameras uses deep learning to predict the *next-best-view* (NBV), maximizing information gain and minimizing occlusion.

2.1.2 Contact-Based Systems

Robots attach sensors directly to leaves or stems to measure physiological and biochemical traits:

- Vision systems segment organs; custom grippers position the sensors.
- One system integrates RGB, hyperspectral, thermal, ToF, and fluorescence sensors onto a robotic arm.

- A 4-DOF arm with deep learning-guided grasping uses a fiber-optic spectrometer and a thermistor to collect leaf reflectance and temperature, and a linear potentiometer to measure stem diameter in maize and sorghum. Predictive models estimate chlorophyll, water, nitrogen, phosphorus, and potassium content.
- Another system uses a SPAD meter on a robotic arm to measure chlorophyll in ornamental plants like Anthurium and Pothos, achieving up to an 82% success rate in leaf probing despite challenges in surface adaptation.

2.2 Outdoor Systems

In the field, plants are stationary and robots must be mobile, resilient to sunlight, wind, and uneven terrain, and capable of autonomous navigation using GPS and local sensors.

2.2.1 Non-Contact Unmanned Ground Vehicles (UGVs)

These platforms use LiDAR, RGB, ToF, near-infrared, or stereo cameras—either fixed or on manipulator arms—to reconstruct 3D geometry and extract traits via image processing and machine learning:

- Measured traits include plant height, leaf angle, area, width, stalk number (in maize, sorghum, sunflower, cabbage), cotton boll count, peanut canopy cover, grape berry size/color, and vineyard volume/yield.
- A compact UGV extracts latent phenotypic features from fused RGB and LiDAR data using machine learning.
- A system with a 6-DOF arm and a 3D sensor measures height and Leaf Area Index (LAI) in maize and sorghum.
- Other platforms use VIS/NIR cameras for wheat NDVI, spectral sensors for biomass and reflectance, or hyperspectral/thermal/LiDAR combinations for canopy closure, temperature, and vegetation indices.

2.2.2 Contact-Based UGVs

These require precise organ grasping and 3D reconstruction:

- A robot uses a 3-DOF arm, a stereo camera, and a penetrometer with a force gauge to measure stem lodging resistance in sorghum.
- Another integrates a leaf spectrometer for full hyperspectral scans, predicting nitrogen content with high accuracy ($R^2 = 0.73$).
- Grasping success varies from 68% to 82%, limited by wind, occlusion, and segmentation errors.

3 Summary of Current Capabilities

- **Crops:** Maize and sorghum dominate due to their economic importance, genetic diversity, and favorable morphology (alternate leaves, central cylindrical stalk).
- **Traits:** Height, width, and volume are the most frequently measured (each in more than 5% of studies), followed by leaf dimensions, reflectance, chlorophyll, temperature, LAI, NDVI, stalk strength, and fruit traits. Architectural traits influence light interception, nitrogen use, and yield.
- **Accuracy:** Height and stem diameter are measured most reliably due to clear geometric signals and linear sensors.
- **Sensing:** Non-contact methods dominate. Sensor fusion (e.g., RGB+depth, RGB+LiDAR, spectral+ToF) improves robustness compared to single modalities.
- **Software:** ROS is the dominant framework for sensor-actuator integration, communication, and navigation.

- **Platforms:** 86% are custom-built; 88% use wheels or wheeled legs for maneuverability and speed; 12% use tracks for traction.

4 Key Challenges

- **Plant Complexity:** Occlusion, deformable structures, intra- and inter-species variation, and variable lighting/backgrounds complicate organ segmentation. Multispectral and thermal signals are sensitive to distance, angle, curvature, and shading.
- **Unstructured Environment:** Limited manipulator workspace, GPS failure under dense canopies, variable light, wind, soil conditions, and dynamic obstacles challenge navigation and control.
- **Software:** Open-source libraries often lack long-term support; real-time constraints cause latency and collision risks.
- **Practical Issues:** Big Data management, limited battery life, fragility under harsh conditions (dust, humidity, heat), high costs, and poor generalizability hinder commercialization.

5 Future Advancements and Applications

5.1 Sensor-Controller Fusion

Combining RGB, LiDAR, and IMU data with hybrid AI (fuzzy logic, reinforcement learning, genetic algorithms) will improve robustness in dynamic environments.

5.2 Internet of Robotic Things (IoRT)

Cloud connectivity enables real-time data streaming, remote monitoring, and multi-robot coordination.

5.3 Renewable Energy

Solar panels and hydrogen fuel cells will extend mission duration in remote fields.

5.4 New Measurable Traits

- **Physiological:** Stomatal conductance, gas exchange, chlorophyll fluorescence, sap flow, and lodging resistance using soft or vacuum grippers and clip-on sensors.
- **Reproductive:** Inflorescence size, kernel traits, fruit quality (sugar, water, firmness) via spectroscopy.
- **Roots:** High-resolution 3D imaging using rotating plant tables and robotic arms.

Deep Reinforcement Learning (DRL) can optimize collision-free trajectories.

5.5 Greenhouse Integration

Stationary robotic arms can complement conveyor-belt-based imaging systems by performing direct chemical and physiological measurements.

5.6 Swarm Robotics

Multiple small robots enable parallel and distributed sensing over large areas with decentralized control.

5.7 UAV-UGV Collaboration

Drones provide aerial mapping and obstacle detection; ground robots perform high-payload proximal sensing.

6 Conclusion

Autonomous agricultural robots have already transformed plant phenotyping by increasing speed, throughput, repeatability, and accuracy across species, environments, and traits. Remaining challenges in sensing, localization, navigation, and data management require interdisciplinary collaboration among plant scientists, engineers, and computer scientists. Advancements in sensor fusion, AI control, renewable energy, IoRT, and multi-robot systems will overcome plant and environmental complexities. With sustained investment, the authors predict that within a decade, intelligent, generic, and affordable phenotyping robots will be fully integrated into breeding programs—monitoring thousands of plants in real time and accelerating the development of productive, resilient, and sustainable crops.

7 References

References

- [1] Atefi, A., Ge, Y., Pitla, S., Schnable, J. (2021).
Robotic Technologies for High-Throughput Plant Phenotyping: Contemporary Reviews and Future Perspectives.
Frontiers in Plant Science, 12, 611940.
<https://doi.org/10.3389/fpls.2021.611940>